

Supplementary Material S4

Complete Stratified Synthesis of the Corpus

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder:
A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements the table in the manuscript presenting the stratified synthesis restricted to the 11 studies of greatest translational relevance (Section 6.3). Here the 31 studies of the complete corpus are reported, with the additional columns of population, architecture and eligibility group that do not fit in the printed version.

The numbering of the # column corresponds to the internal order of the corpus, identical to that employed in Supplementary Material S2, and not to the bibliographic reference numbers of the manuscript. The 3 retracted studies were excluded from the corpus following the editorial status verification described in Section 2.7 of the manuscript and do not appear in this table.

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data provenance	Group
1	Johnson et al.	Emotion recogn.	AffectNet+clinical	ASD children	Ecological	EfficientNet-v2	ACC 67.8%	Clinical	Eligible
2	He et al.	Eye-tracking	Private (N=42)	ASD presch. (n=21)	Clinical	Eye-tracking	r=0.619	Clinical	Eligible
3	Reed et al.	Epidemiological	ALSPAC (N=3,562)	Paediatric	Populat.	Regression	b=-0.18	Cohort	Eligible
4	Lecciso et al.	Therapeutic interv.	Private (N=24)	ASD children	Therapy	OpenFace	g=0.52	Clinical	Eligible
5	Shaik et al.	ASD detection	Kaggle ASD	Children	Lab.	Ensemble CNN	ACC 84%	Web	Eligible
6	Ahmad et al.	ASD detection	Kaggle ASD	Children	Lab.	ResNet50	ACC 92%	Web	Eligible
7	El Rhatassi 2024	Emotion recogn.	FER2013+private	Autistic adults	Lab.	CNN vs ViT	ACC 65-74%	Public general	Contextual ref.
8	Radočaj & Mart.	Emotion recogn.	FERAC	ASD children	Lab.	Swin Transformer	ACC 80%	Partial prov.	Eligible
9	Washington et al.	Emotion recogn.	Clinical+CAFE	Children	Lab./Ecol.	ResNet-152	Bal.ACC 79.1%	Clinical	Eligible
10	Sholikah et al.	Real-time recogn.	FER2013	N=3 ASD	Mobile app	VGG-16	ACC 91%	Public general	Eligible
11	Levy et al.	Wearable system	FER2013	ASD adults	Ecological	CNN+HW	No metrics	Public general	Contextual ref.
12	Sariyanidi et al.	Autism pred.	CHOP clinical	ASD+TD (N=42)	Clinical	Dict.Learning+SVM	Fig.	Clinical	Eligible
13	Hassan et al.	Therapy monit.	DREAM (N=61)	ASD children	Robot th.	RF+XGBoost+ExtraT.	ACC 95.6%	Public curated	Eligible
14	Alsubai	ASD detection tab.	Kaggle screening	12-36m	Screening	Transformer MDNCT	ACC 94.7%	Web	Eligible
15	Rudovic 2018	Affect/therapy	MIT private (N=35)	ASD children	Robot th.	PPA-net	ICC ~60%	Clinical	Eligible
16	Cicceri et al.	Pointing gestures	Clinical private	ASC children	Clinical	4-level framework	mAP 88.6%	Clinical	Eligible
17	Rudovic 2019	Active learning	MIT private (N=35)	ASD children	Robot th.	4 LSTMs+Deep RL	Fig.	Clinical	Eligible
18	Almeida et al.	Serious game	No ML dataset	ASD 6-12	Game	RPG (no ML)	ISO/IEC	N/A	Eligible
19	Alam et al.	ASD detection	Kaggle+UFIID	Children	Lab.	ASD-UANet	ACC 96%	Web	Eligible
20	Hwang et al.	DD screening	Private (N=89)	34-77m	Mobile app	LSTM+2D-FAN	ACC 88%	Clinical	Eligible
21	Aljbori et al.	Emotion recogn.	Kaggle	ASD children	Lab.	3D-CNN+LSTM	ACC 99.82%	Web	Eligible
22	El Rhatassi 2025	Real-time recogn.	CK++Kaggle	Autistic adults	Lab.	CNN×3+LSTM	ACC val. 53%	Mixed	Contextual ref.
23	Ramachandran et al.	Emotion recogn.	Kaggle	Children	Lab.	GWO-GEfficientNet	ACC 96.5%	Web	Eligible
24	Hosney et al.	ASD detection	Kaggle (co-author)	Children	Lab.	YOLOv8+DCNN	ACC 97.2%	Web	Eligible
25	Nagae & Lee	Therapy+EDA	Private (N=11)	DD children	Robot th.	EDA+wavelet	Conf. matrix	Clinical	Contextual ref.
26	Papaioannou et al.	EEG cogn. load	EEG (N=63)	ASD+ADHD	Clinical	MSE+graphs	ANOVA	Clinical	Contextual ref.
27	Dapogny et al.	Facial expression	JEMImE (N=157)	Children 6-11	Clin. game	Rand. Forests+PCRF	ACC ~82%	Public curated	Eligible
28	Leo et al.	Facial prod. ASD	CK++clinical (N=17)	ASD children	Clinical	HOG+SVM+CNN	TP 85% (CK+)	Clinical	Eligible
29	Akin-Bulbul et al.	ET screening	Private ET	Infants 18-36m	Clinical	KNN+features	ACC 81.45%	Clinical	Eligible
30	Xu & Ju	VBM brain	ABIDE (N=203)	ASD 8-12 (n=98)	Clinical	VBM	GMV-ADOS	Public curated	Contextual ref.

(Continued)

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data provenance	Group
31	O’Sullivan et al.	Speaker diarisation	Roche (N=72)	ASD 5–45	Trial	Pyannote+ECAPA	r=0.81	Trial	Contextual ref.

Note. The Group column identifies the membership of each study in one of the two sets defined in Section 2.4 of the manuscript: the 24 eligible studies forming the main synthesis and the 7 contextual reference studies that do not fully meet the eligibility criteria. Only the studies marked as eligible participate in the denominator of the methodological challenges table of the manuscript. The Data provenance column classifies the provenance of the dataset employed according to the criteria documented in the ethical and clinical quality appraisal table of the datasets (Section 2.6 of the manuscript): **Clinical** (collected in a clinical or therapeutic context with institutional oversight), **Public curated** (public with diagnostic validation and consent documented in a primary publication), **Public general** (public without specific orientation to ASD), **Partial prov.** (dataset whose documentation derives solely from third parties who reused it, without a primary publication permitting verification of consent, diagnostic validation or licence), **Web** (collected from web repositories without consent or verified diagnosis), **Mixed** (combination of datasets of differing provenance), and **Cohort** and **Trial** (population cohort and registered clinical trial, respectively). This classification is objectively derivable from the provenance declared by the authors and replaces the aggregate MMAT score as the stratification criterion, in accordance with the methodological justification set out in Section 2.6 of the manuscript. In studies 2 and 30, the total N of the dataset includes a typically developing (TD) control group; the n in parentheses in the Population column indicates specifically the subgroup with an ASD diagnosis. The label Fig. in the Metrics column indicates that the study reports its supervised classification metrics in graphical form only, not numerically extractable: this applies to studies 12 and 17. In study 28 the true positive rate of 85% corresponds to the validation of the system on the public CK+ dataset, composed of typically developing adult participants, and not to its subsequent application to the 17 children with ASD, for which the study reports expressive production scores rather than classification metrics.

Correspondence with the reduced version in the manuscript

The table printed in the manuscript comprises the 11 eligible studies that operate on datasets of clinical, curated public or partial provenance, and that therefore constitute the evidence of greatest translational relevance in the corpus: studies 1, 4, 8, 9, 13, 15, 16, 17, 20, 27 and 28. That subset is defined by dataset provenance and not by clinical domain, which is why it comprises 5 studies from domain D1, 5 from D2 and 1 from D4. The remaining 20 studies appear solely in this document.

Subset	N	Studies (# column)
Direct translational relevance (5 D1, 5 D2, 1 D4)	11	1, 4, 8, 9, 13, 15, 16, 17, 20, 27, 28
Domains D1 and D2 on web sources	3	10, 21, 23
Domain D3, recognition by the child	3	2, 3, 18
Domain D4, screening or diagnosis	7	5, 6, 12, 14, 19, 24, 29
Domain D5, contextual reference	7	7, 11, 22, 25, 26, 30, 31
Definitive corpus	31	

The counts in this table sum to 31 and are mutually exclusive. Study 10 (Sholikah et al.) belongs to domain D1 but employs a general benchmark public dataset, and studies 21 and 23 employ datasets from web repositories, which is why all three fall outside the reduced version despite belonging to a primary evidence domain. Study 20 (Hwang et al.) belongs to domain D4 and does appear in the reduced version as it operates on its own clinical data in a mobile application, in accordance with the provenance criterion declared in the preceding note.

Analytical subsets used in the manuscript

The manuscript operates on several subsets of the corpus according to the research question addressed. Their membership is enumerated below using the numbering of the # column, so that every denominator employed in the manuscript can be traced to the individual studies that compose it.

Analytical subset	N	Studies (# column)
Identifiable facial ML architecture	21	1, 5, 6, 7, 8, 9, 10, 12, 13, 15, 16, 17, 19, 20, 21, 22, 23, 24, 27, 28, 29
Extractable supervised metrics	13	1, 5, 8, 9, 10, 12, 13, 17, 20, 21, 22, 23, 24
Complete numerical values	11	the preceding subset excluding 12 and 17, which report graphically
Primary evidence (domains D1 and D2)	13	1, 4, 8, 9, 10, 13, 15, 16, 17, 21, 23, 27, 28
Direct translational relevance	11	1, 4, 8, 9, 13, 15, 16, 17, 20, 27, 28

These subsets respond to independent criteria and are not mutually exclusive. The facial ML architecture subset is the denominator of the architecture, hyperparameter and class-balancing tables; its membership coincides with the ML base column of Supplementary Material S2. The primary evidence subset and the extractable supervised metrics subset share only eight studies (1, 8, 9, 10, 13, 17, 21 and 23) despite both comprising 13, a numerical coincidence made explicit in the manuscript abstract. The Metrics column of the main table above reports, for each of the 31 studies, the principal figure published by its authors, whatever its nature; several studies outside the extractable supervised metrics subset therefore display accuracy values that are not incorporated into the stratified performance synthesis of the manuscript, since that synthesis is restricted to the enumerated subset.

Interpretive precisions on the scope of reported metrics

Five studies report figures whose scope is not self-evident from the manuscript’s performance table. They are consolidated here so that each value can be read against the exact task, dataset partition and population from which it derives.

Study 1, Johnson et al. (ACC 67.8%). The figure corresponds to classification into seven emotion categories by an EfficientNet-v2 model trained and evaluated on the AffectNet dataset; the same publication also reports 64.65% for the eight-category variant. The 48 participants recorded in the Participants column belong to a complementary clinical study in which users assessed enhanced expression images, and did not generate the 67.8% figure. The dataset is recorded as clinical in provenance on account of that complementary component.

Study 9, Washington et al. (Bal.ACC 79.1%). The figure corresponds to the subset of the paediatric CAFE dataset restricted to images with human labelling agreement at or above 60%, designated Subset A in the manuscript. On the complete CAFE set the same model reports 66.9% balanced accuracy. Both figures derive from the same model and differ only in the evaluation subset.

Study 10, Sholikah et al. (ACC 91.0%). The figure corresponds to the evaluation of a VGG-16 architecture on the FER-2013 dataset across seven expression categories, with an equal F1 value. The three participants with ASD recorded in the manuscript took part in a subsequent usability trial assessing interest in learning expressions, and did not generate the 91.0% figure.

Studies 12 and 17, Sariyanidi et al. and Rudovic 2019 (Fig.). Neither publication reports its supervised classification metrics numerically. Sariyanidi et al. present autism-versus-neurotypical classification accuracy solely as a performance curve against the number of expression components; the only numerical accuracies stated in that publication, 73% and 32%, correspond to classification using head-movement signals alone and not to the proposed method. Rudovic 2019 presents accuracy and F1 in graphical form only. Both remain within the subset of 13 studies reporting supervised metrics, so the number of studies with numerically extracted values is 11.

Study 28, Leo et al. (TP 85%, CK+). The true positive rate corresponds to validation of the system on the public CK+ dataset, composed of typically developing adult participants, where the reported figures are TP 141/167 = 85%, FP 2/20 = 10%, TN 18/20 = 90% and FN 25/167 = 15%. For the subsequent application to the 17 children with ASD the study reports expressive production scores contrasted against annotation by three psychologists, and not classification metrics; there is no control group of typically developing children in that phase.

Caveat on translational relevance

As documented in Section 7.1 of the manuscript, none of the 31 studies of this corpus evaluates session-to-session longitudinal tracking. The designation of direct translational relevance applied to 11 of them indicates proximity of object, that is, that their outcome variable corresponds to facial expression or to affect and engagement during sessions, and proximity of data provenance; it in no case indicates proximity of design or evidence of demonstrated therapeutic effectiveness.